

Antimicrobial Effectiveness of Photodynamic Therapy, 5% Sodium Hypochlorite and 2% Chlorhexidine Gluconate in Root Canal Treated Teeth: A Clinical Study

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List of Abbreviations

CHO: Chlorhexidine gluconate

MB: Methylene blue

NaOCl: Sodium hypochlorite

PDT: Photodynamic therapy

PS: Photosensitizer

S1: First microbiological sample

S2: Second microbiological sample

Introduction

The residual root canal infection is the major reason for failure of endodontic treatment.^{1–3} Disinfection with sodium hypochlorite (NaOCl) and chlorhexidine gluconate (CHO)

has a favourable treatment outcome. However, these root canal irrigants do not completely remove the bacteria present in the infected root canals.

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Chlorhexidine gluconate solutions of different concentrations have been recommended as endodontic irrigants^{4,5} and dressings.⁶⁻⁸ NaOCl and Chlorhexidine both have distinct antibacterial activity. In a study irrigation done either with 1% NaOCl or 2% CHO both effectively reduce the total number of viable bacteria (*streptococci* and *E. faecalis*) in root canal treated teeth with apical periodontitis. However, there was no significant difference in both the groups.⁹

Results of previous research have been inconclusive when comparing the antimicrobial effectiveness of 2% NaOCl and 2% chlorhexidine solutions.¹⁰⁻¹² The main drawbacks of conventional root canal treatment are that it's a time-taking procedure and teeth became weak due to enlargement of the root canal.¹³ Still, the complete disinfection of root canal is a challenge. Because of bacterial drug resistance, there is a need of both for new irritants and new techniques to disinfect root canal space.

Photodynamic therapy (PDT) was used to activate a nontoxic photosensitizer, to release nascent oxygen which causes damage to the membrane and DNA of microbes.¹⁴ So, PDT is recommended as a substitute or an additive to currently used disinfection procedures.^{15,16} In an in vivo study it was used successfully for the removing of multi-drug resistant microorganisms.¹⁷ Bacterial culture methods are commonly used and are easily available to evaluate the antibacterial effectiveness of the different irrigations.

The objective of this in vivo study was to compare the effectiveness of PDT, NaOCl and CHO in disinfecting infected teeth that had previously undergone root canal treatment. The null hypothesis was that there would be no difference in elimination of microbes between the PDT, NaOCl and CHO.

Materials and Methods

Selection of the Patients

This trial was a triple blind randomized clinical trial of three groups. This was performed from September 2016 and July 2017 at King George's Medical University, Lucknow. This clinical study was approved from the University Ethical Committee (Ref. no. 77th ECMII-B IMR faculty/p15). Standards of the Helsinki Declaration were followed.¹⁸

With reference to the findings of the study by Asnaashari et al.,¹⁹ sample size estimation was done using OpenEpi software (Atlanta, GA, USA) considering power of the study = 80%, confidence interval = 95%. The sample size was 18 in each group.

In a total of 54 patients (25 men and 29 women, age between 20 and 70 years, mean age 38 years), each tooth

was endodontically treated with apical periodontitis. Patients were chosen from a pool of patients who were referred for root canal re-treatment.

Inclusion and Exclusion Criteria

Patients with previously root canal treated teeth with apical periodontitis were taken for the study. Teeth with a periapical index of 3 or more were incorporated in the study.

In all the cases, root canal treatment was done more than 3 years earlier. There was no overextended obturation in all cases. On clinical examination, all teeth had intact coronal restoration. Chosen teeth had no gingival recession and no periodontal pocket more than 4 mm. Before the treatment was initiated, these patients had not taken any antibiotics for a period of at least one month. Patients with any condition that were immunocompromising in nature along with conditions such as diabetes mellitus, chronic infections, rheumatoid arthritis were excluded from the study. Patients on immunosuppressive therapy, pregnant and nursing women were also excluded from the study. The study included the teeth that could not be easily isolated.

Study protocol was explained to all the patients and written informed consent were taken. Root canal treated teeth with chronic apical periodontitis either on maxillary or mandibular anterior teeth was the diagnosis in all of the cases. These patients were randomly divided into three groups based on the intervention used. We have used block randomization method for this purpose. The same practitioner performed the whole clinical protocol. Microbiological samples were collected by another practitioner to avoid any bias.

Endodontic Procedures

Pre-Sampling Procedure for All Groups

Rubber dam was used to isolate the teeth from the oral cavity. All coronal restorations, carious defects and supra-gingival plaque was removed. Throughout the endodontic treatment and sample collection care was taken to maintain asepsis. To constrain the microbial assessment to a single ecological area, a single root canal was sampled in each case. After accessing the canal with Gates-Glidden burs (Dentsply Maillefer, Ballaigues, Switzerland), up to coronal 3rd of the canal. Further, Rotary ProTaper retreatment files were used to remove the gutta-percha and sealer (ProTaper Universal Retreatment, Dentsply Maillefer, Ballaigues, Switzerland). No chemical solvent of any kind was used to remove the gutta-percha.²⁰ After removing the gutta-percha, sterile saline solution was used to fill the canal and pulp chamber. Saline was not allowed

to overflow from the cavity. Entire saline solution was absorbed from the root canal and pulp chamber by using paper points. Paper points are highly absorbent and they absorb all the fluid present in the root canal. They are easy to take microbial samples from the canals, while the endodontic files and Gates-Glidden drills are used only when we require the dentin chips for sample.

Paper points were transferred to the Eppendorf tube (Eppendorf, Hamburg, Germany) comprising anaerobe transport media (thioglycolate broth, Anaero GRO, Hardy Diagnostics, USA). Paper points were inserted into the canal to the full working length (which was assessed using a preoperative radiograph) and kept for 1 minute. This sample was the first microbiological sample (S1) sent for microbial isolation. Now at this stage, patients were randomly distributed to different groups according to disinfection protocol.

Group 1: Disinfection with 5% Sodium Hypochlorite

After the first sample (S1), root canal was irrigated with a copious amount of 5% NaOCl for 5 minutes using a 5-mL syringe and 30-G needle, which was placed 2 mm short of the working length. Root canal preparation was completed with hand files. After each treatment, the canals were thoroughly irrigated with 2 mL 5% sodium thiosulfate followed by 2 mL normal saline solution, to neutralize the effect of NaOCl solution.

Then, according to the working length, which was assessed with a preoperative radiograph, a sterile paper point was inserted into the canal to take the microbial sample. The paper point was kept in the canal for 60 seconds to completely absorb NaOCl. Subsequently, the paper points were transferred to 1 mL 0.85% sterile saline in the Eppendorf tube and vortexed vigorously for 1 minute. This was the second microbiological sample (S2) of this group.

Group 2: Disinfection with 2% Chlorhexidine

Using a 5-mL syringe and 30-G needle, which was positioned 2 mm short of the WL, the root canal was irrigated with 5 mL 2% chlorhexidine for 60 seconds. After each treatment with chlorhexidine, the root canals were irrigated with 2 mL 5% sodium thiosulfate to neutralize the effect of chlorhexidine solution followed by 2 mL normal saline prior to take microbial sample.

For microbial sampling, a sterile paper point was inserted into the canal up to the working length and kept in the canal for 60 seconds. Subsequently, the paper points were transferred to 1 mL 0.85% sterile saline in the Eppendorf tube and vortexed vigorously for 1 minute. This is the second microbiological sample (S2) of this group.

Group 3: Disinfection with PDT

A FotoSan Device (FotoSan; CMS Dental, Copenhagen, Denmark) at 628 nm was used throughout the experiment. For PDT on teeth, a long-tapered tip (endodontic tip) was used. The apical size of the tip was 500 μ m and a 0.03 taper in the apical part; it could be inserted into the canals up to 3 mm short of the working length and guide the light to the apical parts.

A watery solution with concentration of 100 μ g/mL of toluidine blue O (TBO; Sigma-Aldrich, St. Louis, MO), a thiazine dye of the quinone-imine family, was prepared and injected into the canal with a sterile single-use endodontic syringe (VMK-Endo needle; Vedefar, Dilbeek, Belgium). A sterile endodontic K-file moved gently up-down in the canal to eliminate air bubbles (ISO 15, stainless steel; Dentsply Maillefer). After this, PDT procedure was carried out. Sixty seconds after inoculation, irradiation treatment was performed with LED lamp for 60 seconds. After irradiation, the root canals were thoroughly irrigated with 2 mL 5% sodium thiosulfate followed by 2 mL normal saline to neutralize the solution remaining in the root canal prior to take the microbial sample. Then the microbiological sample was taken with sterile paper points. Subsequently, the paper points were transferred to 1 mL 0.85% sterile saline in the Eppendorf tube and vortexed vigorously for 1 minute. This was the second microbiological sample (S2) of this group.

Apical preparation was completed with nickel-titanium hand files according to the root canal anatomy and initial diameter of the canal. Each canal was dried using sterile paper points, 5% sodium thiosulfate solutions (Sigma-Aldrich) was used to neutralize any residual NaOCl or CHO and photosensitizer.²¹

Lateral compaction technique with AH plus sealer (Dentsply, De Trey, Konstanz, Germany) was used to obturate the root canals in a one visit root canal re-treatment. Surplus gutta-percha and root canal sealer was eliminated from the pulp chambers. The access cavities were sealed with composite resin (Gaenial, GC, Tokyo, Japan) after a base of glass ionomer cement (Fuji IX GP Extra, GC, Tokyo, Japan) was placed.

Microbiological Procedures

Within 1 hour of the sampling, the microbiological samples were forwarded to the Department of Microbiology, in an anaerobic jar with Gas Pack (BBL, Becton Dickinson, Heidelberg, Germany).²²

A temperature of 37 $\pm$ 1 $^{\circ}$ C for 48 hours was used for the incubation of the plates. Light microscopy at 400 $\times$ was used to examine microbial growth after incubation. Colony-forming units (CFU) was used for microbial

counting, and the average percentage reduction of viable microorganisms was determined.

Conclusion

Many studies concluded that *E. faecalis* was commonly present bacteria in failed root canal treated cases. NaOCl in its various concentrations and 2% CHO are used to treat the *E. faecalis* infection in root canals and have promising results. Application of PDT as an alternative to traditional endodontic treatment might lead to a rapid reduction of pathogens without causing any damage to the healthy soft tissue. This study also suggests that PDT can achieve better infection control in the apical region of canal. Clinical and radiological outcome in all three groups is almost similar except PDT is more effective than NaOCl and CHO.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

Ethical Approval

Ethical committee of King George's Medical University UP Lucknow approved the study.

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Patient Declaration of Consent (If In-vivo Study/ Case reports)

Consent of patients was taken before commencement of study.

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